# Frank — AMC Medal Drill L2 / L3 stretch

25 Jun 2026 | Same two topics, Medal intensity. These are **harder than the real-paper ceiling**: trial-and-error will not finish them — you must reason structurally. Whole-number answers; show working.

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**1.** **[L2]** [7 marks] What is the smallest three-digit number with all of these properties: divisible by 3; contains the digit 6; has no digit 0; has all different digits; has a digit sum divisible by 5?

**2.** **[L2]** [7 marks] What is the smallest three-digit number with all of these properties: divisible by 6; contains the digit 5; has no digit 0; has all different digits; has exactly 1 odd digit?

**3.** **[L3]** [7 marks] In how many ways can you choose 3 different numbers from 3 to 22 so that their sum is a multiple of 3?

**4.** **[L3]** [7 marks] In how many ways can you choose 3 different numbers from 3 to 26 so that their sum is a multiple of 4?

**5.** **[L3]** [8 marks] How many four-digit numbers have all of these properties: divisible by 13; has all different digits; has no digit 0; has exactly 3 odd digits; has a digit sum divisible by 3; contains the digit 2?

**6.** **[L3]** [8 marks] You write the 250 numbers 3, 6, 9, ... up to 750. How many times do you write the digit '0'?

L2 = an in-range question with a hidden condition (don't stop early). L3 = above the real-paper ceiling: the search space is too big to list, so count by structure. Every item is a code-verified twin of a real AMC anchor family.